

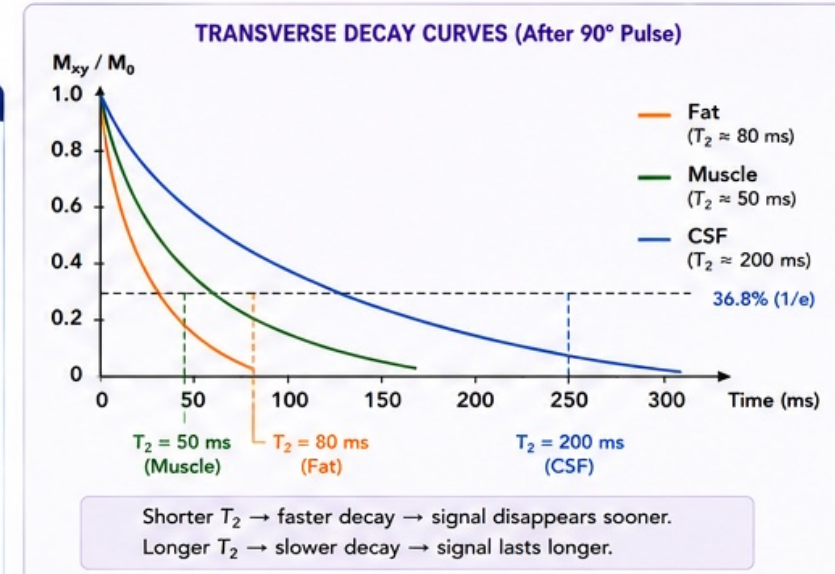
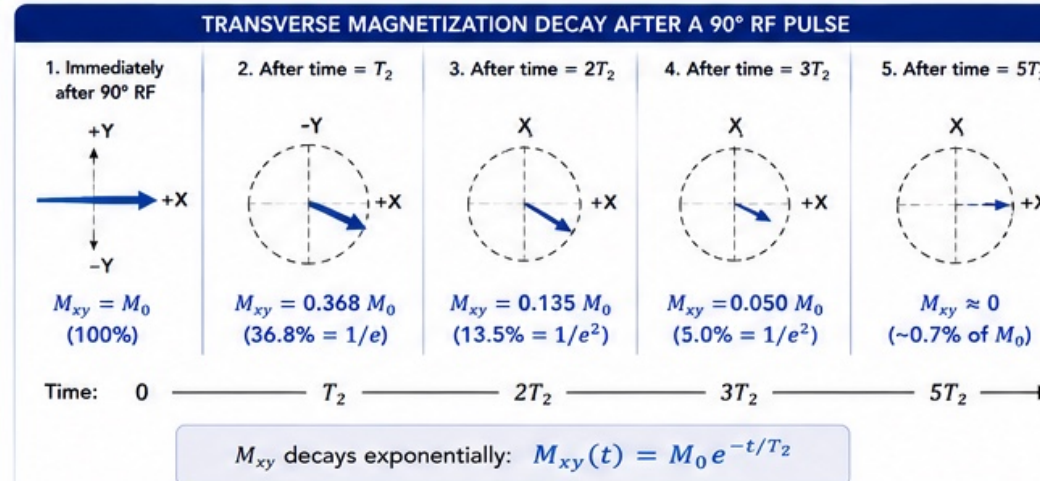
1.10 T2 DECAY (TRANSVERSE RELAXATION)

After excitation, transverse magnetization (M_{xy}) decays to zero with time constant T_2 .

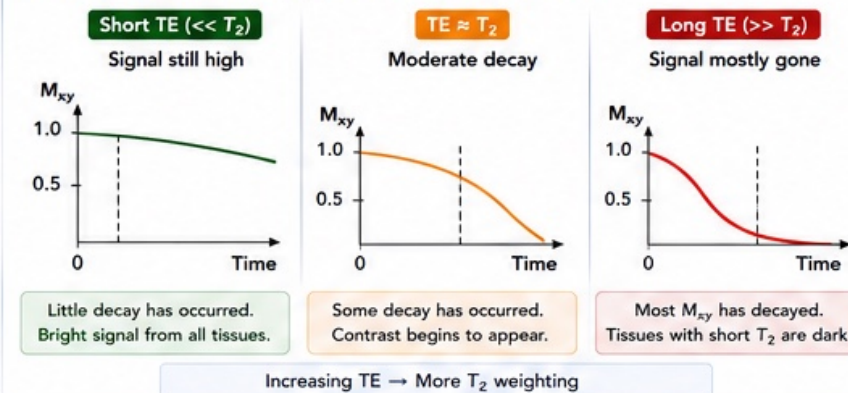
T_2 (spin-spin relaxation) is the process by which spins lose phase coherence with each other in the transverse plane.

This causes the transverse magnetization (M_{xy}) to decay exponentially, even though longitudinal magnetization (M_z) is unchanged during this time.

The time constant T_2 is the time it takes for M_{xy} to decay to **36.8% (1/e)** of its initial value.



EFFECT OF ECHO TIME (TE) ON TRANSVERSE SIGNAL



TYPICAL T2 VALUES AT 1.5T (approx.)

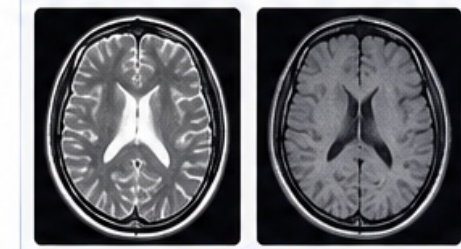
Tissue	T_2 (ms)
Fat	60 – 100
Liver	40 – 60
Skeletal Muscle	45 – 60
Gray Matter	80 – 110
White Matter	70 – 100
CSF	150 – 250+

Values vary with field strength, sequence, and vendor.

CLINICAL RELEVANCE (T_2 WEIGHTING)

- ✓ Tissues with short T_2 (fat, muscle, gray/white matter) lose signal quickly $\rightarrow$ appear dark on long TE images.
- ✓ Tissues with long T_2 (fluid, edema, CSF) retain signal $\rightarrow$ appear bright.
- ✓ Used in SE, FSE/TSE, STIR, T_2^* , and many 2D/3D sequences.
- ✓ Excellent for detecting pathology with increased water content (edema, inflammation, fluid).

Example: T2-Weighted Brain Images



Long T_2 (CSF) $\rightarrow$ bright on T_2 images
Short T_2 (white matter, fat) $\rightarrow$ dark on T_2 images

ENGINEERING INSIGHT

- T_2 decay is caused by microscopic field inhomogeneities.
- These local field variations make spins precess at slightly different rates, causing dephasing.
- T_2 is independent of T_1 . They are separate relaxation processes.
- Stronger B_0 uniformity $\rightarrow$ less dephasing $\rightarrow$ longer T_2^* .



FIELD SERVICE TIP

- Poor shimming shortens T_2^* (and can reduce apparent T_2).
- Check and optimize shimming, especially for sequences using long TE.
- Gradient eddy currents and system instability can also accelerate decay.
- Verify receiver bandwidth settings—higher BW $\rightarrow$ shorter T_2^* .



COMMON MISCONCEPTION

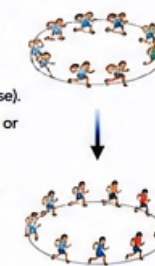
“If a tissue has a long T_2 , it will always be bright.”

Reality: Signal also depends on TR, TE, proton density, T_1 recovery, and sequence design (e.g., STIR suppresses fat even though fat has a long T_2).

Context is everything!

REAL-WORLD ANALOGY

Imagine a group of runners (spins) running around a circular track.
At first they are side-by-side (in phase).
Over time, each runner slows down or speeds up slightly.
They drift apart (dephase), and the group no longer runs together $\rightarrow$ the signal fades.



KEY TAKEAWAY

- T_2 decay is the loss of phase coherence in the transverse plane.
- M_{xy} decays exponentially with time constant T_2 .
- Shorter T_2 $\rightarrow$ faster decay $\rightarrow$ dark on long TE images.
- Longer T_2 $\rightarrow$ slower decay $\rightarrow$ bright on long TE images.

